Few-Shot Multispectral-Hyperspectral Image Collaborative Classification With Feature Distribution Enhancement and Subdomain Alignment

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***Abstract*— With the development of observation technology, multispectral (MS) images of large scenes are easy to obtain, but the low spectral resolution limits their classification ability. Moreover, the collection of training samples is difficult and time-consuming, and limited labeled samples are a challenge for the precise classification of large-scene MS images. This article attempts to use hyperspectral (HS) images with limited labels to help classify MS images of large scenes, so as to achieve better classification results. To solve this problem, a few-shot MS-HS image collaborative classification method combining feature distribution enhancement (FDE) and subdomain alignment is proposed. Specifically, a residual 3-D convolution network embedded with a 3-D FDE module is designed to improve the diversity of the feature distribution extracted by the network and increase the generalization ability of the model under the few-shot condition. Furthermore, the local domain alignment between the source and target domains is achieved by subdomain alignment, which better aligns the categories in the source domain and the target domain, and achieves the distribution alignment of the subdomains. In addition, the feature bias adjustment (FBA) module is introduced in the test phase to correct the bias of the MS image feature representation, and to alleviate the cross-domain problem to some extent. The few-shot learning (FSL) is applied in the source and target domains to learn better feature mapping. The results of comparative experiments on three datasets show that the proposed method is superior to the most advanced method in the case of limited labeled samples.**

***Index Terms*— Classification, domain adaptation, few-shot learning (FSL), hyperspectral (HS) image, multispectral (MS) image.**

I. Introduction

HYPERSPECTRAL (HS) images and multispectral (MS) images are two types of images that are widely used in the field of remote sensing. They play a key role in applications such as agriculture and environmental management [1], image classification [2], [3], target detection [4], and change detection [5], and have important application values in both militaries [6] and civilian fields [7]. Among them, accurate classification of remote sensing images is the basis and prerequisite for many applications and plays an important role in the field of remote sensing. HS images consist of hundreds of continuous bands, and each pixel can obtain a continuous spectral curve with richer spectral information. HS images can reflect the subtle spectral properties of the ground objects to be observed in detail, and it is easier to distinguish different ground objects. However, due to equipment limitations, the imaging range of HS images is very narrow. Furthermore, the quantity of HS satellites is limited, and the revisiting period of these satellites is long. As a result, HS satellites are not conducive to short-term and large-scale observation of the area of interest. With the development of remote sensing technology, MS satellites are increasingly available, and MS data can be obtained more easily. In contrast, the MS image has a large imaging range, but the low spectral resolution makes it impossible for MS remote sensing to obtain highly recognizable spectral features of surface materials, which limits its ability to classify ground objects. Therefore, if MS images and HS images can be combined in a complementary way, more accurate classification of MS images of large scenes can be achieved. However, the spectral mismatch problem arises due to the inconsistent spectral feature dimensions of the MS and HS images. Moreover, since limited labeled samples account for an extremely small proportion of all samples in large-scene MS images, it is difficult to achieve the precise classification of large-scene MS images with limited labeled samples. Therefore, one of the most pressing issues in remote sensing image classification is how to effectively leverage the benefits of both MS and HS images by transferring the superior spectral of HS images to the large-scale classification task of MS images and achieving precise classification with limited labeled samples.

In fact, few-shot MS-HS collaborative classification can be regarded as a cross-modal learning problem [8]. This problem can be solved by transfer learning and few-shot learning (FSL).

Since MS images and HS images have different dimensions, heterogeneous transfer learning methods can be employed in research. Shekhar et al. [9] proposed a shared domain dictionary learning method, which can not only maintain the manifold structure of the data but also learn a discriminative

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dictionary when learning feature transformation. Tian et al. [10] achieved structure preservation and spatial alignment by transferring feature representation and model knowledge from the source domain. Zhu et al. [11] advocated the deep subdomain adaptation network to adapt the subdomain for transferring. Li et al. [12] introduced a new semi-supervised distribution-regularized matrix factorization method to learn a dictionary and new instance representations by exploiting the correlation between instances and features. In the field of remote sensing cross-modal learning, Hong et al. [13] proposed the common subspace learning (Cospace) method, which learns shared subspaces linearly from HS-MS of overlapping regions to achieve better classification results. Based on Cospace, Guo et al. [14] carried out domain adaptation between the large-area HS image reconstructed by MS and the real small-area HS image, so as to improve the classification ability of the image. Hong et al. [15] proposed a multimodal deep learning (MDL) framework that can combine different fusion strategies to achieve pixel-level classification tasks on heterogeneous data. In [16], a generative adversarial network (GAN) driven cross-modal network is devised for semantic segmentation in complex large-scale urban scenes. Zhao et al. [17] presented a novel cross-scene HS image classification framework, featuring a convolutional neural network and a transformer-based multistage feature extractor. This framework effectively captured feature representations from both domains while minimizing the gap between source and target domains through two-stage alignment. In [18], discriminative least squares and spatial constraints are combined to achieve multi HS collaborative classification. Hong et al. [19] designed a novel cross-modal semi-supervised learning network, called X-ModalNet, which combines three designed modules to achieve classification of MS images and SAR images with the assistance of HS images. Fang et al. [20] proposed the confident learning-based domain adaptation (CLDA) method combining adversarial and assertive learning techniques to achieve favorable classification results. But the above methods require the source and target domains to share the same class space, and the ability to transfer will decrease when new classes appear. There are also some deep learning-based methods that perform transfer learning by fine-tuning the network [21], [22]. Fine-tuning-based methods do not require the source domain and target domain classes to be consistent, but require the distribution of the two domains to have a certain level of similarity and require a requisite number of target domain samples during fine-tuning to achieve better results. The *n* negative transfer may occur in transfer learning when there is a significant dissimilarity between the source and target domains and only a limited number of labeled samples.

In recent years, FSL has garnered significant attention due to its capability to achieve high classification accuracy with only a small number of training samples [23], [24], [25], [26]. It mainly includes three types, which are model-based [27], optimization-based [28], and metric-based [29], [30]. Metric-based FSL has attracted the attention of many scholars due to its simplicity and practicality. In actual scenarios, because remote sensing data is more complex than natural images, it is more difficult to label remote sensing data. In order to address the shortage of labeled samples, many scholars who inspired by FSL have proposed FSL methods for solving HS image classification. Liu et al. [31] developed a deep FSL (DFSL) method to classify HS images by learning the metric space through deep residual networks. On this basis, considering the cross-domain problem, Li et al. [32] proposed a new deep cross-domain FSL (DCFSL) method, which adopts a conditional adversarial domain adaptation strategy to achieve domain distribution alignment while performing FSL. But adversarial learning through symmetry will affect the class boundary of the target domain, thereby affecting the effect of feature learning [33]. Based on the prototype network, Yu et al. [34] combined the calibrated aggregation network and calibrated metric learning for HS few-shot classification. Wang et al. [35] proposed the heterogeneous FSL (HFSL) method, which initialized the network model weights on the mini-ImageNet dataset and then fine-tuned the FSL on the HS images. Similarly, Li et al. [36] proposed a novel few-shot classification framework with self-supervised learning (FSCF-SSL) to achieve HS images classification of new classes with the assistance of natural images. This method combines geometric transformations and contrastive learning to learn transferable spatial meta-knowledge to improve the classification ability of HS images. Liu et al. [37] combined supervised contrastive learning and FSL into an end-to-end network for few-shot HS images classification and proposed an improved FSL prototype contrastive learning network. Yu et al. [38] enhanced the ability to distinguish nodes of different categories by using graph convolutional neural networks (GNNs) to measure the similarity of features and instances separately. Based on GNN, Wang et al. [39] realized heterogeneous HS image classification through the spatial importance strategy and the conditional domain adversarial strategy. Zhang et al. [40] proposed graph information aggregation cross-domain FSL (Gia-CFSL) and achieved better results by performing domain alignment from two aspects of graph features and graph distribution levels in GNN. Wang et al. [41] introduced the kernel method to extract HS image features, and designed a classifier based on cosine similarity to achieve few-shot classification. Starting from the metric criterion, Xi et al. [42] applied the class-covariance metric-based FSL (CMFSL) method for HS images classification, which realized the classification with Mahalanobis distance and task-adapted class covariance estimation. Sun et al. [43] used the Earth mover's distance as the FSL metric to reduce irrelevant background interference and achieve HS few-shot classification.

Although the above cross-domain few-shot classification methods have achieved good results, the following challenges currently exist in the field:

1) Prevailing cross-domain FSL classification methods primarily focus on scenarios involving HS and HS images. Compared to these methods, the spectral mismatch in cross-domain FSL between HS and MS images is more heavily and more prone to overfitting due to the lack of samples and the large domain bias, thus reducing the generalization ability of the model. Therefore, reducing overfitting to improve the generalization of cross-domain FSL models is one of the research priorities.

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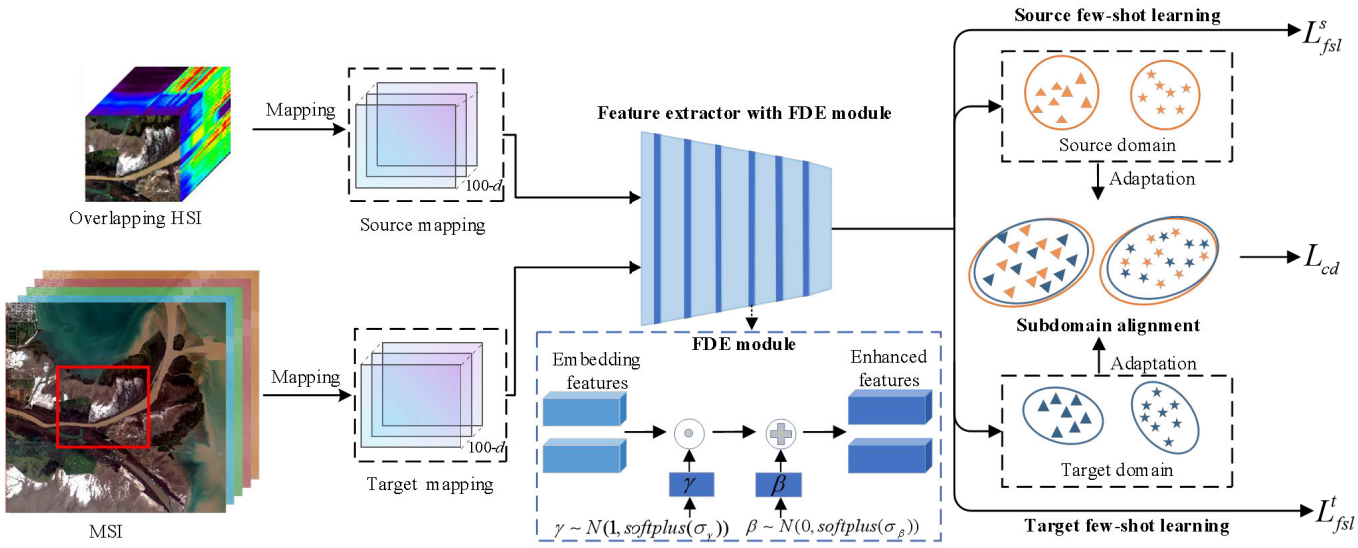


Fig. 1. Proposed model for the classification task. It consists of four parts that are mapping layer, feature extraction network, FSL, and subdomain alignment.

2) Most of the cross-domain FSL methods rely on global feature alignment, which can only align marginal distributions, regardless of the distribution between sub-domains in two domains, and do not consider a large number of discriminative local features in the learning process. Forcibly aligning global features may lead to poor performance when subdomain features are mapped to the wrong classes.

In order to solve these challenges, this article proposes a few-shot MS-HS collaborative classification model, which is mainly composed of a feature extraction module embedding the feature distribution enhancement (FDE) module, FSL, subdomain alignment, and feature bias adjustment (FBA) module. First, the feature extraction module is composed of multiple 3-D residual convolution blocks and 3-D FDE, which are used to generate a variety of data feature distributions. Second, the source domain and target domain FSL are utilized to learn the metric space. At the same time, the subdomain alignment module is used for domain alignment to align classes within the domain. In the test phase, the FBA module is introduced to correct the image features in the target domain and further improve the classification performance. The main contributions of this article can be summarized as follows:

1) To the best of our knowledge, the work is among the first to realize large scenes cross-domain few-shot classification of MS images with the help of partial HS images.
2) In this article, the generalization capability of the model is enhanced by both FDE and feature bias reduction. The 3-D FDE module embedded in the feature extraction stage is designed to simulate the feature distribution generated in different fields, so that the feature extractor is more suitable for feature extraction in the target field. The FBA module is introduced to reduce the feature shift between the source domain and the target domain, so as to achieve better classification results.
3) To avoid feature mismapping caused by global alignment in most cross-domain few-shot methods, the local alignment method is employed to adapt the features of the source domain and the target domain in the subdomain, and the alignment of local and global features is realized in MS-HS data through the constraint of local maximum mean discrepancy (LMMD).

The rest of the article is organized as follows. Section II introduces the proposed method in detail. Section III presents the specific implementation of experimental data and experimental results, and Section IV concludes the article.

II. Proposed Algorithm

The proposed method solves the problem of collaborative classification of MS-HS images within the same observation scene. Its core aim is to transfer the excellent spectral resolution capabilities of HS images under the common coverage area to large-scene MS images, thus enhancing the classification abilities of MS images. This section provides a comprehensive description of the proposed method.

A. Method Overview

As shown in Fig. 1, the framework of this method mainly includes four parts, which are the feature mapping layer, feature extraction, FSL, and subdomain alignment. The mapping layer utilizes two mapping layers to first map the sample features of the source and target domains to the same dimension, and then the feature extraction extracts features from the samples with the same dimension. The proposed method utilizes the residual 3-D convolutional neural network (CNN) for feature extraction and uses the sample distribution enhancement module to simulate the distribution of different features during the training phase, thereby improving the generalization ability of the model. In the feature extraction part, the gray part represents the feature extractor embedded in the FDE module. The source domain and the target domain carry out the FSL of the source domain and the target domain respectively through the FSL. At the same time, the difference between the source and target domains is reduced by the LMMD to

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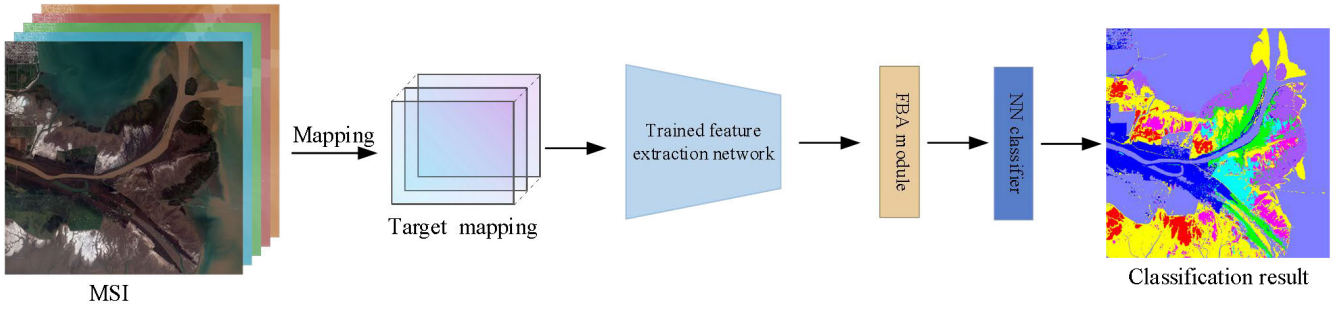


Fig. 2. Flowchart in the testing phase.

achieve cross-domain subdomain alignment. Through LMMD and FSL, the feature extractor maps the same kind of features in the source and target domains as closely as possible, and the different classes of features as far away as possible. The overall loss function of the method is:

$$L = L_{\text{fsl}}^{s} + L_{\text{fsl}}^{t} + L_{\text{cd}} \tag{1}$$

where L_{fsl}^{s} and L_{fsl}^{t} are the loss of FSL respectively, and L_{cd} is the loss of local maximum mean difference.

In the test phase, to further reduce the bias between the source domain and the target domain samples, the FBA module is introduced. As illustrated in Fig. 2, the target domain data is initially mapped to the same dimension as the source domain data using the mapping layer. Subsequently, the trained deep residual network extracts discriminative features with a narrow intraclass distance and a broad interclass distance. Finally, after reducing the bias between the target domain and the source domain by the feature bias module, the target domain data are classified by the NN classifier.

B. Feature Mapping Layer

Due to the inconsistent spectral resolution of MS and HS images, a feature mapping layer is used to map source domain HS images and target domain MS images to the same feature dimension, so that the MS-HS images can utilize the same feature mapping network structure for deep feature extraction.

The mapping layer converts the channel number of the source domain (HS images) and the target domain (MS images) into 100 through 2DCNN, which is expressed as follows:

$$S' = S \times P \tag{2}$$

where $S \in R^{9\times9\times ch}$ is the input data, ch is the number of HS and MS bands, $P \in R^{ch\times100}$ is the mapping matrix, and $S' \in R^{9\times9\times100}$ is the size of the mapped data. In the mapping transformation, $ch \times 100$ parameters need to be learned. Specifically, there are 295×100 parameters for GF5 HS data and 4×100 parameters for GF1 MS data.

C. Feature Extraction Network

The feature extractor depicted in Fig. 1 is illustrated in Fig. 3, which mainly consists of residual 3-D CNN and the FDE module.

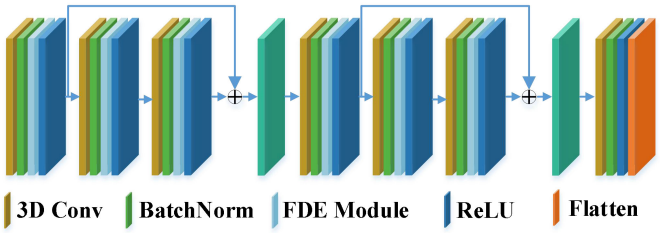


Fig. 3. Structure of the feature extraction network.

1) 3DCNN and Residual Learning: The HS data is a 3-D block, and 2-D convolution usually performs dimensionality reduction processing when analyzing 3-D data, which will cause information loss. In order to make full use of the spectral-spatial features of HS data, the 3DCNN is utilized for spectral-spatial feature extraction. The 3-D convolution performs convolution operations on three degrees of freedom to obtain more features than 2DCNN. The 3-D convolution at the position (x, y, z) can be calculated as follows:

$$v_{ij}^{xyz} = f\left(b_{ij} + \sum_{m}\sum_{o=0}^{O_i-1}\sum_{q=0}^{Q_i-1}\sum_{r=0}^{R_i-1}\alpha_{ijm}^{oqr}v_{(i-1)m}^{(x+o)(y+q)(z+r)}\right) \tag{3}$$

where α_{ijm}^{oqr} denotes the value of the position (o, q, r) associated with the mth feature map, R_i, O_i, and Q_i are the spectral dimension, height, and width of the convolution kernel, respectively. $f(\cdot)$ is the activation function and b_{ij} is the bias.

For the CNN network, the more layers of the network, the richer abstract features of different levels can be extracted, but simply increasing the depth of the network will easily lead to gradient disappearance and explosion. Deep residual learning methods have been proposed to address the degradation problem of deep networks. Assuming that the network input is x, the learned features are $H(x)$, and the residual can be defined as:

$$F(x) = H(x) - x \tag{4}$$

where the input of the next layer is $F(x) + x$. When the residual $F(x) = 0$, the input of the next layer is x. By implementing identity mapping and overlaying the input information onto the final output, a skip connection is formed across several layers, effectively alleviating the gradient disappearance or explosion

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phenomena. Residual learning can preserve the depth of the deep network and has the advantages of the shallow network to avoid the degradation problem. This residual learning structure is implemented through the forward neural network and shortcut link, where the shortcut link is equivalent to simply performing the same mapping, without generating additional parameters, and without increasing computational complexity. The entire network can still be trained through end-to-end backpropagation.

2) 3-D Feature Distribution Enhancement Module: For MS-HS collaborative classification tasks, it is necessary to make full use of limited labeled HS data. However, MS images and HS images acquisition sensors are different, and there are differences in spectral resolution and spatial resolution, which lead to differences in their feature distributions. In the training phase, the metric function may overfit the HS feature distribution, resulting in the weak generalization ability of the measurement function in the MS test stage. In order to solve this problem, inspired by [27], the 3-D FDE module is proposed, which uses an affine transformation to simulate different distributions of HS features during training, and different feature distributions are generated to improve the generalization ability of the model to MS images. Let $\mathbf{y} \in R^{C\times H\times W\times D}$ be the feature extracted by the feature extractor. In this method, the feature distribution extension layer is connected to the back of the batch normalization layer in the feature extractor, and the intermediate feature activation is added to enhance the diversity of data feature distribution in the training stage, which is expressed as follows:

$$\widehat{y}_{C,H,W,D} = \gamma_C \times y_{C,H,W,D} + \beta_C \tag{5}$$

where $\widehat{y}_{C,H,W,D} \in \widehat{\mathbf{y}}$ and $y_{C,H,W,D} \in \mathbf{y}$, the sampling scaling item γ and bias item β from the Gaussian distribution can be expressed as follows:

$$\gamma \sim N\big(1, \text{softplus}(\sigma_\gamma)\big)\beta \sim N\big(0, \text{softplus}(\sigma_\beta)\big) \tag{6}$$

where $\sigma_\gamma \in R^{C\times 1\times 1\times 1}$ and $\sigma_\beta \in R^{C\times 1\times 1\times 1}$ represent the standard deviation of the Gaussian distribution used for affine transformation. The enhancement of the feature distribution can be completed by (4). After the FDE module is utilized during the training phase, the diversity of the feature distribution in the training data can be enhanced, thereby improving the generalization ability of the model for MS images and achieving better classification results.

D. Few-Shot Learning

The spectral-spatial features of the source domain and the target domain are subjected to feature mapping and FSL simultaneously. The Euclidean distance between unlabeled samples and labeled samples of each class is calculated respectively in the source domain and target domain for FSL. However, since there are very few samples in the target domain, FSL cannot be formed. Therefore, for the target domain part, accumulation noise is added through only a few labeled samples, and FSL is performed after data expansion. In the process of FSL, the training method of the episode is commonly utilized, where each round of training involves multiple training tasks.

Within the training set of either the source or target domain, a collection of N classes is selected at random and M labeled samples are chosen per class, resulting in a dataset of $M \times N$ entries. The samples and labels of the support set can be expressed as $S_s = \{(x_i, L_i)\}_{i=1}^{N\times M}$. As the support set, a batch of K samples different from the query set is extracted from the remaining data of N classes as the query set of the model. The samples and labels of the query set can be expressed as $Q_s = \{(x_j, L_j)\}_{i=1}^{N\times K}$. The predicted probability output of samples in the query set can be expressed as follows:

$$\Gamma\big(\hat{L}_j = z|\hat{x}_j \in Q\big) = \frac{\exp\big(-\text{ED}\big(f_\theta(\hat{x}_j), c_k\big)\big)}{\Sigma_{k=1}^{C'}\exp\big(-\text{ED}\big(f_\theta(\hat{x}_j), c_k\big)} \tag{7}$$

where $\text{ED}(\cdot)$ represents the Euclidean distance, the predicted probability output is the Euclidean distance calculated by the softmax function. c_k is the cth class prototype, which is the feature of the kth class mapping in the support set, and $\hat{x}_j$ is the query set samples. C' is the number of different samples in one set, and f_θ represents the mapping layer and map extraction network. Therefore, the classification loss in source and target domains can be expressed as:

$$L_{\text{fsl}}^s = E_{S_s,Q_s}\left[-\sum_{(x,L)\in Q_s}\log\Gamma_\theta(L = z|x)\right] \tag{8}$$

$$L_{\text{fsl}}^t = E_{S_t,Q_t}\left[-\sum_{(x,L)\in Q_t}\log\Gamma_\theta(L = z|x)\right] \tag{9}$$

where z represents the true label, Q_t and Q_s are the support set and query set from the source domain dataset, respectively.

E. Subdomain Alignment

In order to reduce the interdomain difference between the target domain and the source domain, this method adopts the local maximum mean difference to adjust the local distribution of the source and target domains. Adversarial-based domain adaptation methods usually face the problem of slow and difficult convergence, and the loss is complicated. Inspired by the DSAN network [11], LMMD is utilized to reduce the difference between domains, and classes are used as the basis for division. The samples of the same label are placed in a subdomain for the subdomain alignment. In the field of transfer learning, many deep methods adopt the maximum mean discrepancy (MMD) to evaluate the distribution difference between the source domain and the target domain. The MMD finds a continuous function g so that the mean difference has the maximum value. However, these methods mainly focus on the adaptation of the global distribution and do not consider the degree of adaptation between two domains of the same category. In our method, the source domain D_s and the target domain D_t are divided into multiple sub-domains according to the class information c. Since the target domain samples have very few labels, the target domain samples use pseudo-labels. $E_c[\cdot]$ represents the mathematical expectation of the category, and LMMD can be defined as follows:

$$D_{\text{H}}(j,k) = E_c\left\|E_{j^{(c)}}\left[\phi\left(\mathbf{x}^s\right)\right] - E_{k^{(c)}}\left[\phi\left(\mathbf{x}^t\right)\right]\right\|_{\text{H}}^2 \tag{10}$$

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where $\mathbf{x}^s$ and $\mathbf{x}^t$ are the features of the source and target domains, respectively, and $j^{(c)}$ and $k^{(c)}$ represent the distribution of $D_s^{(c)}$ and $D_t^{(c)}$. Samples belonging to the same class exhibit a higher correlation. Equation (9) can be used to express the MMD distance of the same class, and it can be minimized to adapt to the distribution of similar samples between the source domain and the target domain, achieving distribution alignment within subdomains. Assuming that there are n_s labeled samples in the source domain and n_t unlabeled samples in the target domain, which obey the distributions j and k, respectively. Through unbiased estimation and mathematical techniques, the above formula can be expressed as follows:

$$\hat{d}(j,k) = \frac{1}{C}\sum_{c=1}^{C}\left[\sum_{i=1}^{n_s}\sum_{j=1}^{n_s} w_i^{sc} w_j^{sc} k\left(\mathbf{y}_i^t, \mathbf{y}_j^t\right) - 2\sum_{i=1}^{n_t}\sum_{j=1}^{n_t} w_i^{sc} w_j^{tc} k\left(\mathbf{y}_i^s, \mathbf{y}_j^t\right) + \sum_{i=1}^{n_t}\sum_{j=1}^{n_t} w_i^{tc} w_j^{tc} k\left(\mathbf{y}_i^t, \mathbf{y}_j^t\right)\right]. \tag{11}$$

Equation (10) illustrates that LMMD measures the Hilbert–Schmidt norm between kernel-averaged embeddings of empirical distributions in relevant subdomains of the source and target domains, while accounting for varying sample weights. Among them, the weight w_i^c of each sample belongs to the class c, that is, w_i^{sc} and w_j^{tc} represent the weight of the sample $\mathbf{x}_i^s$ and $\mathbf{x}_j^t$ in class c, respectively, and the calculation formula can be expressed as follows:

$$w_i^c = \frac{L_{ic}}{\sum_{(\mathbf{x}_j,\mathbf{y}_j)\in\mathcal{D}} L_{jc}} \tag{12}$$

where L_{ic} is the label of the samples, and the source domain data has labels $\mathbf{L}_i^s$ and w_i^c can be directly calculated. Since the target domain data has few labels, the output of the deep neural network is a probability distribution, which can represent the probability of assigning $\mathbf{x}_i^t$ to each C class. However, directly using the label prediction method is prone to errors and leads to negative transfer. Therefore, probabilistic prediction is utilized to generate soft labels for target domain data, and LMMD is integrated with the deep neural network, training the network by constraining LMMD to minimize it. Therefore, the local domain adaptation module can promote the distribution of two domains of the same class to be closer, thereby improving the ability of the model to classify MS images.

F. Feature Bias Adjustment Module

Since FSL requires the model to be able to quickly adapt to new tasks, [26] has shown that the importance of different channels for different datasets is different. Due to differences in MS and HS image sensors, there are differences in data distribution. Inspired [26], a channel transformation function is introduced to adjust the average channel amplitude on the target task and vary the importance of the different channels.

The channel transformation function mainly acts on the average channel amplitude, suppresses channels with large amplitudes and amplifies low-amplitude channels to a large extent. Let the feature of the lth channel be defined as the lth dimension of $\mathbf{y}$. Let $\mathbf{x}$ denote an image and $f_\theta(\cdot)$ denote a feature extractor learned from training on the training set, and $\{y_i\}_{i=1}^d$ is the set of all d channels. The channel transformation function is defined as follows:

$$\Psi_k(\varphi) = \begin{cases} \dfrac{1}{\ln^\tau\left(\frac{1}{\varphi}+1\right)}, & \varphi > 0 \\ 0, & \varphi = 0 \end{cases} \tag{13}$$

where $\tau > 0$ is a hyperparameter. At test time, each channel of the image features is transformed by this function, i.e.,

$$\Psi_k(\mathbf{y}) = (\Psi_k(y_1), \ldots, \Psi_k(y_d)). \tag{14}$$

Since the function can only use non-negative eigenvalues, the rectified linear unit (ReLU) activation function is adopted. Moreover, this method uses the channel feature bias module to further reduce the distribution bias between the MS data in the target domain and the source domain, and achieve better classification results.

III. Experimental Results and Discussions

In this section, we verify the effectiveness of the proposed algorithm through three HS-MS datasets of experiments and result analysis. Our code will be released at: https://github.com/GCBin/FDESA.

A. Data Description

The method proposed in this article aims to solve the HS-MS collaborative classification problem. To assess the effectiveness of the proposed model, three different HS-MS datasets are employed in this study. The particulars of each dataset are described as follows:

1) *University of Houston:* The University of Houston (UH) dataset comprises both a Houston HS image as well as a simulation-generated Houston MS image. The HS image in the Houston dataset is acquired by the ITRES CASI-1500 sensor and provided by the 2013 IEEE GRSS Data Fusion Competition. The spatial resolution of the image is 2.5 m, the data size is 349 × 1905, and it contains 144 bands with a spectral range from 364 to 1046 nm. The MS image in the Houston dataset is generated by downsampling the HS data using the Sentinel-2 spectral response function (SRF), which includes ten bands. During the training phase, the study utilizes the overlapping regions of HS and MS in the dataset, while non-overlapping regions are reserved for testing. In the source domain, a fixed number of 50 samples are used for training in the overlapping area. If classes have less than 50 samples in the overlapping area, all available samples are used for training. For example, the water class has 30 samples for training, while the Parking Lot 2 class has 20 samples. The target domain data randomly selects ten samples from each category in the overlapping area to participate in the training process.

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TABLE I
INFORMATION FOR THE UH DATASET

Class No.	Color	Class Name	TR in Source	TE
1		Healthy Grass	50	1186
2		Stressed Grass	50	1165
3		Synthetic Grass	50	647
4		Tree	50	1194
5		Soil	50	1192
6		Water	30	295
7		Residential	50	1205
8		Commercial	50	1194
9		Road	50	1175
10		Highway	50	1177
11		Railway	50	1168
12		Parking Lot 1	50	1183
13		Parking Lot 2	20	449
14		Tennis Court	50	378
15		Running Track	50	610
Total			3908	14218

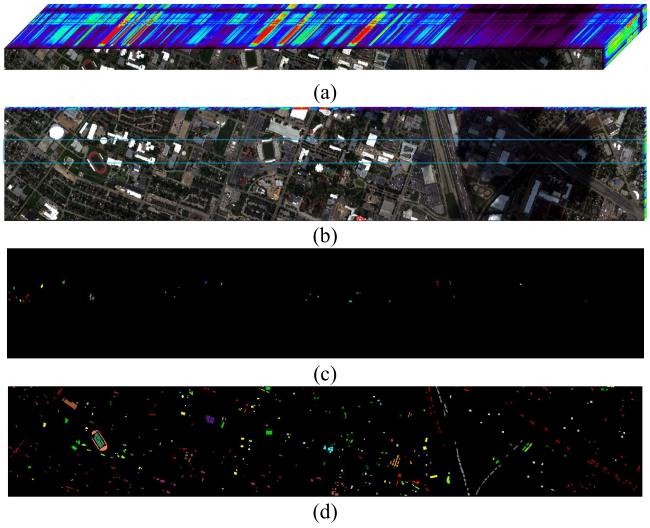


Fig. 4. UH dataset. (a) Hyperspectral false color composite image. (b) Multispectral false color composite image. (c) Training labels. (d) Test labels.

TABLE II
INFORMATION FOR THE YRE_1 AND YRE_2 DATASETS

Class No.	Color	Class Name	TR in Source	TE	
				YRE_1	YRE_2
1		Reed	50	46555	98275
2		Tamarix	50	26109	55139
3		Tidal Flat Reed	50	20667	43891
4		Salt Marsh	50	18952	40126
5		Suaeda Salsa	50	25477	53874
6		Mudflat	50	110314	232701
7		Water	50	266686	562243
8		Spartina Alterniflora	50	44882	94479
Total			400	559642	1180728

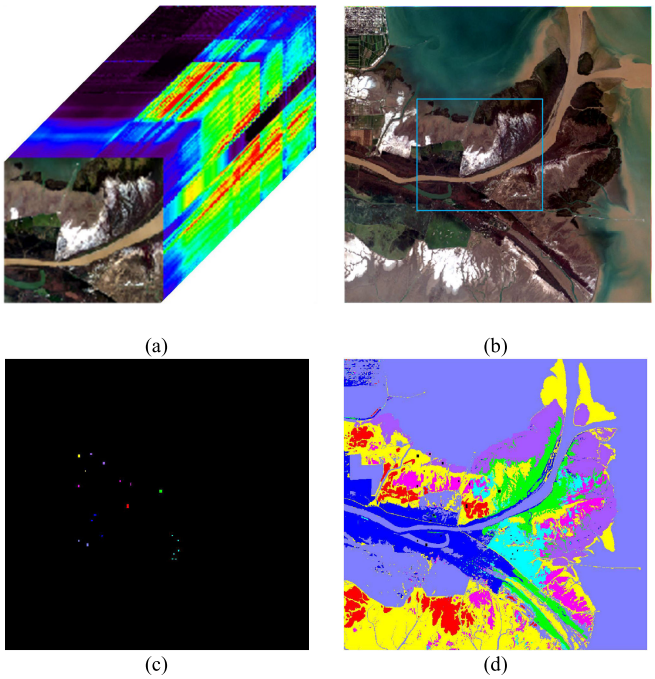


Fig. 5. Yellow River Estuary_1. (a) Hyperspectral false color composite image. (b) Multispectral false color composite image. (c) Training labels. (d) Test labels.

Fig. 4 presents false-color images of both the training and test sets of the UH dataset, and the specific classes of samples and quantities are shown in Table I.

2) *Yellow River Estuary_1:* Yellow River Estuary_1 (YRE_1) dataset includes GF5 HS images, as well as simulated GF1 MS images. The HS data in the dataset comes from the GF5 satellite, including 330 bands, taken in the Yellow River Estuary area on November 1, 2018, with a spatial resolution of 30 m and a spectral range of 0.390–2.513 μm. The MS data are sampled from GF5 image by the SRF of GF1, including four bands with a spectral range from 0.45 to 0.89 μm. The size of MS is 734 $\times$ 763 and the size of overlapping regions of HS and MS is 278 $\times$ 309. Similar to the UH dataset, the source domain selects 50 fixed samples from the common coverage area in HS, while the target domain randomly selects ten samples to train in the common coverage area before testing on the non-common coverage area. Fig. 5 displays the training and test figure of GF1 and GF5 images. The dataset includes eight classes, and specific settings are presented in Table II.

3) *Yellow River Estuary_2:* Yellow River Estuary_2 (YRE_2) dataset also contains a pair of HS and MS images, in which the HS data is the same as the YRE_1 dataset. The MS data with 1066 $\times$ 1108 pixels captured by WFV GF1 was taken on October 31, 2018, and comprises four bands with a spatial resolution of 16 m. Table II provides detailed settings for the training samples. The false color images, training labels, and test labels are similar to those present in the YRE_1 dataset.

B. Experimental Setting

In order to verify the effectiveness of the proposed method, the method is compared with the other eight methods, which are SVM, 3DCNN, CLDA, DCFSL, Gia-CFSL, HFSL, CMFSL, and FSCF-SSL. Among them, SVM is a non-deep method, which is based on the principle of structural risk minimization and can solve the problem of small samples.

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3DCNN is a classic deep-learning classification method, which can effectively extract the spectral-spatial features of samples. The H-3DCNN and M-3DCNN methods only involved a single domain in the experiment. H-3DCNN employs HS training and HS testing, while M-3DCNN uses MS training and MS testing. Both H-3DCNN and M-3DCNN are used as the basic comparison method in the experiment. This article aims to demonstrate that the proposed HS-MS collaborative classification method can achieve better results than the M-3DCNN method for MS classification, and reach or exceed the performance of the H-3DCNN method. CLDA is an adversarial domain adaptation method, which integrates instance-level confidence learning and biclassifier domain adaptation techniques to attain cross-domain classification. Except for CLDA, 3DCNN, and SVM, other methods are advanced cross-domain few-shot classification methods. Among them, DCFSL is the first method to perform cross-domain few-shot classification, utilizing FSL in both the source and target domains, as well as conditional domain discriminator, to learn an effective metric space. Gia-CFSL and CMFSL are improved methods based on DCFSL. Specifically, the Gia-CFSL method combines FSL with graph information aggregation to perform feature-level and distribution-level cross-domain graph alignment. CMFSL designed a spectral prior-based refinement module, which thoroughly utilizes the spectral-spatial information at high and low frequency scales, and determines the class labels based on the Mahalanobis distance. HFSL initially employs source domain samples to learn the initialization weights of the model. Subsequently, it fine-tunes the network using target domain data to achieve cross-domain few-shot classification. FSCF-SSL initially trains the model on the source domain, then transfers part of the model trained on the source domain to the target domain, and finally fine-tunes it in the target domain using the limited number of labeled samples. Both FSCF-SSL and HFSL methods utilize all training sets from mini-ImageNet as the source domain, maintaining consistency with [35] and [36]. For all few-shot classification methods, ten labeled samples are randomly selected from each class as supervised samples in the target domain, and the rest are used as test samples. Except for HFSL and FSCF-SSL, other few-shot classification methods utilize the fixed 50 samples for training in the source domain. In order to eliminate the influence of random sampling and ensure the reliability of the experimental results, all experiments were carried out ten times. We try our best to ensure the fairness of the experiment, and refer to the settings in the original papers to select the best hyperparameter settings for comparison.

In our experiment, the settings are the same as [32], [40], and [42]. In each episode, the size of the MS-HS input cube is 9×9, and the model uses ADAM to optimize the parameters. In order to train more fully, the number of training iterations is set to 10 000, which is enough to learn the prototype of each class. All convolutional layers and linear layers were normalized using Xavier normalization, and the learning rate is set to 0.001, which is a suitable condition for the network to achieve stable convergence. For each N-way K-shot meta-task, N is the number of classes in the target domain. When classifying the YRE_1 and YRE_2 datasets, the number of sampled classes is set to 8, and for the UH dataset, the number of sampled classes is set to 15, respectively. K represents the number of samples of each class in the support set. In each set, the number of query samples N_Q is set to 19, and K is set to 1, that is, only one labeled sample is randomly selected in each set to participate in model training. The more samples the query sample N_Q contains, the better the few-shot class can be simulated in the target domain [32], [40]. In the test phase, the embedded features of the target domain are obtained through the trained feature extractor, and the nearest neighbor classifier is employed to compare the distance between the query embedded features and the class prototype, so as to perform classification. In this article, the overall accuracy (OA), average accuracy (AA), and Kappa coefficient are used to compare the classification performance of different methods.

C. Experimental Results and Analysis

1) Compared With Other Methods: The method proposed in this article is compared with related methods, and the experimental average accuracies are shown in Tables III–V. For the convenience of comparison, the highest accuracy of each row is indicated in bold. Based on the results, it is evident that our method exhibits superiority over the others. The following is a specific analysis for different datasets. Table III displays the classification performance of these methods on the UH dataset. Compared with other comparative methods, the method in this article achieves the best. Furthermore, the OA of the proposed method is approximately 1% higher than the second-highest performing method, FSCF-SSL. The FSCF-SSL method extracts low-level features in different directions from natural images through self-supervised learning with geometric transformation and contrast learning. This process significantly enhances the extraction of spatial features from HS images, resulting in better performance compared to Gia-CFSL. The Gia-CFSL method utilizes the graph distribution information propagation mechanism to map data at both the feature level and the distribution level, thereby achieving better results than the CMFSL method. CMFSL method leverages the Mahalanobis distance for measurement, and makes full use of the high-frequency and low-frequency spectral-space information, resulting in better performance compared to the DCFSL method which employs symmetric adversarial learning for domain adaptation. The HFSL method not only transfers the source domain information to the target domain, but also learns the discriminative features of the target domain, which is better than DCFSL. Comparatively, the CLDA method outperforms the 3DCNN-M, indicating that the domain adaptation method can also improve the MS classification ability. However, due to the limitation of samples, the effect of CLDA is not as good as the cross-domain few-shot classification methods. Among all methods, SVM has the worst classification performance. This is due to the limited number of samples, which makes SVM prone to overfitting. And it only uses spectral information, so the effect is not good. In contrast, 3DCNN exploits the spatial-spectral information at the same time and outperforms the SVM method. It can also be seen from the table that the

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method based on cross-domain FSL is superior to the traditional deep classification method 3DCNN, which illustrates the effectiveness of FSL and cross-domain adaptation. Among them, the result of 3DCNN-H is higher than that of 3DCNN-M, indicating that HS images are more discriminative than MS images. In cross-domain FSL, the resolution of HS images can be transferred to MS images through cross-domain adaptation, thereby improving the classification ability of MS images. It is worth noting that the classification accuracy of most categories of the proposed method is higher than other methods, which means that subdomain alignment and the FDE module can help the network to make more accurate decisions, and the FBA module can further improve classification accuracy. In order to compare the classification performance of the above methods more intuitively, the classification map on the target domain of the dataset is given. Fig. 6 illustrates the resulting map generated by the proposed method on the UH dataset is notably smoother compared to other methods. Especially class 5 and class 3 both achieve better classification results, which verifies the effectiveness of the proposed method.

Table IV shows the classification results of the YRE_1 dataset. The superiority of our method is evident as it achieves the best results, with OA, AA, and Kappa being 2.46%, 1.84%, and 3.29% higher than the suboptimal FSCF-SSL method, respectively. Similar to the results of the UH dataset, it can be seen that the cross-domain FSL method is superior to the CLDA, 3DCNN, and SVM methods in the case of a limited number of samples, indicating that using FSL can learn a better metric space so that different classes can be more clearly distinguished. The classification result map is shown in Fig. 7. It can be seen from the classification diagram that due to the lack of spectral information in MS image, the features extracted by SVM and 3DCNN lack distinctiveness, leading to a significant number of misclassified pixels in MS images with significant noise. What is more, most of the Reed was misclassified as the Tamarix in the CLDA method. In FSCF-SSL method, most Suaeda Salsa is misclassified as Mudflat. In the HFSL method, the Mudflat was seriously misclassified into Suaeda Salsa, which is very different from the truthmap in Fig. 5. In GiaCFSL and CMFSL methods, most of the bare soil below the Yellow River estuary data was mistakenly divided into Suaeda Salsa, and in the DCFSL method, part of the bare soil below was mistakenly divided into Tidal Flat Reed. It can be seen that the classification result of the proposed method is the closest to the truthmap and achieves the best classification result. Table V presents the classification accuracy of the YRE_2 dataset. The experimental results demonstrate that the proposed method in this article achieves the highest classification performance. For the majority of categories, our method ranks either first or second in classification accuracy, which shows the effectiveness of our proposed approach. The classification accuracy of all categories of the proposed method is higher than that of the baseline method 3DCNN-M, indicating that no negative transfer has occurred. The classification accuracy of the cross-domain few-shot classification method is higher than that of the cross-domain adaptation method CLDA, which shows the effectiveness of FSL. Compared to other cross-domain few-shot classification methods, the proposed method increases the diversity of feature distribution in the feature, and further reduces the feature distribution deviation in the target domain. In terms of domain alignment, the proposed method further aligns locally and globally the source and target domain through LMMD, thus greatly reducing the classification error. The classification results of the YRE_2 dataset are shown in Fig. 8. Since the MS images of the dataset are not obtained by HS simulation, the results are more convincing. It can be seen from Fig. 8 that, similar to the YRE_2 dataset, the classification result map obtained by the method proposed in this article is most similar to the ground truth map in Fig. 5, such as classes Reed, Salt Marsh, and Mudflat. What is more, the pixels on the edge of the classification map are less than other methods to misclassify the samples, such as Spartina Alterniflora and Salt Marsh, which are more clearly at the edge of the classification map than other methods. It shows that this method can also extract discriminative feature information for similar boundary pixels, and can transform the original MS image into a more discriminative embedded space, thus obtaining better classification results.

2) Impact of Overlapping Regions of MS-HS Image: This article evaluates the performance of the proposed method in two cases: when the overlap area of HS and MS images is reduced and when the overlapping area is within the boundary region. For single-domain methods, such as SVM, 3DCNN-H, and 3DCNN-M, samples within the overlapping area are randomly selected for training. For the HFSL and FSCF-SSL methods, natural images are still used for training, and samples in the overlapping area are randomly selected during fine-tuning. The specific settings for the samples are shown in Tables VI and VII.

As shown in Table VI, to validate the classification performance with a smaller overlapping area, the ratio of HS image area to MS image area is further reduced to 1:12. Using the same experimental setup as before, 50 samples are employed for training. Since training samples of Tidal Flat Reed are less than 50, it is trained with 20 samples. As shown in Table VIII, each method runs ten times to derive the average value as the final result. Compared with Table V, it can be observed that the classification performance degrades when the overlapping region area is smaller. This is mainly because the number of samples of the third category in the training samples is small and the feature diversity is low, resulting in a decrease in classification accuracy. However, the method proposed in this article still has the best performance compared with other methods.

As shown in Table VII, in order to verify the classification of the coverage area within the boundary region, this article conducts an experiment using the lower boundary area of the MS image and the HS image for the overlapping coverage area. The experimental results are displayed in Table IX. Compared with Table V, it can be seen that compared with the boundary area as a common coverage area and the central area as the coverage area, the classification performance of the method mentioned has not declined. Furthermore, in comparison to other methods, the method mentioned still yields the best results, thus confirming the effectiveness of the method.

3) Impact of Different Numbers of Labeled Samples: In order to verify the performance of the method under

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TABLE III
CLASSIFICATION RESULTS ON THE UH DATASET

Methods	SVM	3DCNN-H	3DCNN-M	CLDA	DCFSL	CMFSL	HFSL	Gia-CFSL	FSCF-SSL	Ours
Healthy Grass	66.15	84.09	80.83	84.81	70.69	**94.31**	92.47	85.60	89.43	93.44
Stressed Grass	87.30	93.61	96.51	82.55	94.73	85.54	69.31	96.02	95.28	**96.79**
Synthetic Grass	99.56	98.69	80.49	98.96	99.28	96.97	100	95.27	99.13	**100**
Tree	84.85	62.80	42.46	98.15	**94.53**	82.81	79.85	85.37	93.90	87.69
Soil	99.11	96.67	94.80	98.61	95.89	98.63	97.84	97.75	97.41	**98.87**
Water	95.56	93.65	87.93	86.46	90.43	91.25	**95.69**	87.69	91.25	89.20
Residential	38.71	59.52	53.81	87.09	70.41	57.76	**85.74**	62.47	46.64	66.91
Commercial	38.98	49.51	54.05	41.64	58.65	72.07	56.17	**68.49**	23.82	60.07
Road	86.17	53.42	53.01	**66.53**	43.95	45.57	58.48	58.37	78.44	58.81
Highway	46.18	68.45	66.31	59.41	73.98	60.64	**89.06**	70.82	84.77	66.88
Railway	1.30	67.30	51.82	39.90	63.76	**95.71**	82.26	74.30	78.40	64.42
Parking Lot1	7.20	53.48	62.38	56.93	56.33	34.12	65.66	47.34	**72.27**	70.05
Parking Lot2	1.74	55.99	47.93	9.26	85.47	92.46	20.47	95.31	80.38	**92.95**
Tennis Court	69.62	97.37	99.98	97.96	95.08	98.11	98.02	92.29	98.32	**98.53**
Running Track	98.77	97.23	91.07	98.81	96.97	98.32	99.73	97.12	99.61	**99.75**
OA	60.03	72.42	68.27	72.28	76.12	77.01	76.91	78.03	78.95	**80.04**
AA	61.41	75.45	64.29	73.79	79.34	80.28	79.38	80.95	81.87	**82.96**
Kappa	0.5571	0.7022	0.6571	0.7114	0.7420	0.7483	0.7469	0.7627	0.7724	**0.7843**

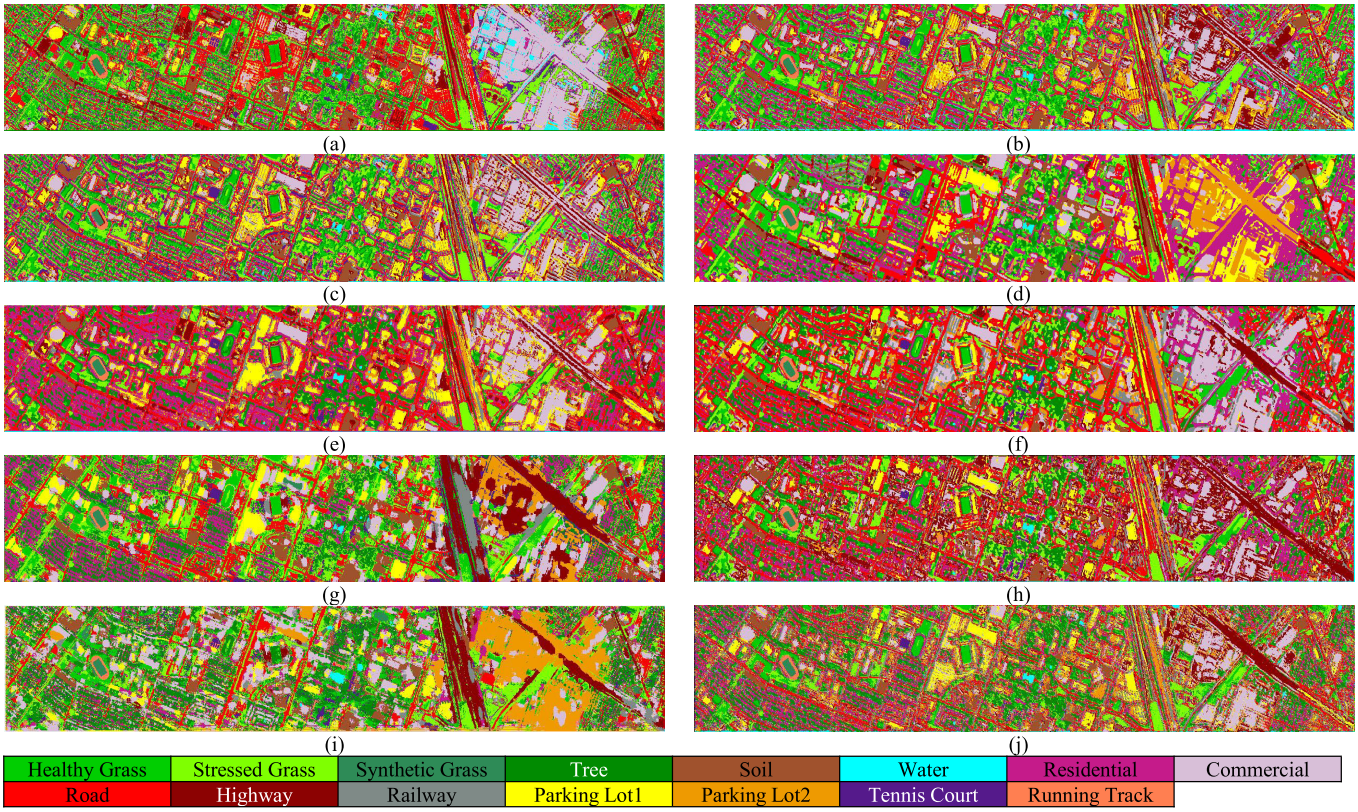


Fig. 6. Classification results on UH dataset. (a) SVM. (b) 3DCNN-H. (c) 3DCNN-M. (d) CLDA. (e) DCFSL. (f) CMFSL. (g) HFSL. (h) Gia-CFSL. (i) FSCF-SSL. (j) Ours method.

varying quantities of labeled data, the proposed algorithm and related cross-domain few-shot classification algorithms, such as HFSL, DCFSL, CMFSL, Gia-CFSL, and FSCF-SSL are experimented with and analyzed when the number of labeled samples in the target domain is reduced. Specifically, the target domain samples with 1, 3, 5, 7, and 10 labels are randomly selected as input for all algorithms in this study. The experiments in this article are repeated ten times, and the OA on the three sets of datasets is shown in Fig. 9. As the number of labeled samples increases, the OA of all classification methods usually increases as well. Across the three datasets, the proposed method consistently demonstrates good performance, thereby affirming its ability to adapt to changes in the number of labeled samples.

D. Alignment Performance

In order to further verify the effectiveness of the proposed method, in the cross-domain FSL methods, the test data in the target domain samples of the three datasets are projected to the 2-D feature space to show the t-distributed stochastic neighbor embedding (t-SNE) map. The experimental results are shown

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TABLE IV
CLASSIFICATION RESULTS ON THE YRE_1 DATASET

Methods	SVM	3DCNN-H	3DCNN-M	CLDA	DCFSL	CMFSL	HFSL	Gia-CFSL	FSCF-SSL	Ours
Reed	16.20	56.35	33.65	48.52	56.29	35.37	**64.15**	29.26	69.22	56.77
Tamarix	**75.66**	72.29	52.83	75.32	48.76	82.20	37.35	66.23	32.07	50.03
Tidal Flat Reed	33.71	38.43	50.54	35.67	30.19	40.35	**62.85**	48.82	65.01	45.12
Salt Marsh	90.62	89.17	63.14	94.45	80.23	**96.50**	97.51	91.85	98.49	90.36
Suaeda Salsa	**73.88**	72.87	48.93	66.34	40.77	67.69	68.94	56.10	23.96	56.13
Mudflat	18.41	28.88	36.58	60.24	51.85	42.23	30.91	59.84	59.76	**60.26**
Water	79.89	85.80	81.09	71.13	85.83	89.11	89.17	88.67	86.50	**89.86**
Spartina Alterniflora	46.99	83.10	47.93	73.93	83.22	70.74	83.20	73.97	82.83	**83.92**
OA	58.02	69.19	61.10	68.49	70.34	71.40	71.01	72.96	73.34	**75.80**
AA	54.42	65.86	51.84	65.70	59.64	65.52	66.64	64.34	64.72	**66.56**
Kappa	0.4554	0.5894	0.4815	0.588	0.5993	0.6114	0.6029	0.6349	0.6361	**0.6696**

TABLE V
CLASSIFICATION RESULTS ON THE YRE_2 DATASET

Methods	SVM	3DCNN-H	3DCNN-M	CLDA	DCFSL	CMFSL	HFSL	Gia-CFSL	FSCF-SSL	Ours
Reed	9.30	56.35	52.13	65.05	58.60	38.85	56.62	58.36	57.16	**60.42**
Tamarix	65.08	**72.29**	34.48	41.17	47.16	71.64	36.89	37.42	62.37	41.24
Tidal Flat Reed	27.33	38.43	27.97	36.75	25.87	48.61	**68.21**	47.68	54.71	40.35
Salt Marsh	87.85	89.17	80.68	98.38	82.95	96.16	**96.95**	92.43	96.07	94.09
Suaeda Salsa	44.07	**72.87**	48.20	45.73	47.01	57.93	50.51	52.71	35.91	50.95
Mudflat	66.79	28.88	55.03	66.41	47.74	44.25	36.59	60.07	56.82	**67.07**
Water	74.46	85.80	77.25	72.19	86.64	86.88	86.92	87.42	85.68	**87.95**
Spartina Alterniflora	40.80	83.10	73.30	80.10	80.98	72.95	82.12	64.65	83.63	**82.71**
OA	62.26	69.19	65.42	68.57	70.14	70.22	70.81	72.56	73.01	**75.09**
AA	51.95	65.86	56.13	63.22	59.62	64.66	64.35	62.59	66.54	**65.60**
Kappa	0.5009	0.5894	0.5425	0.5899	0.5971	0.6043	0.6056	0.6282	0.6369	**0.6574**

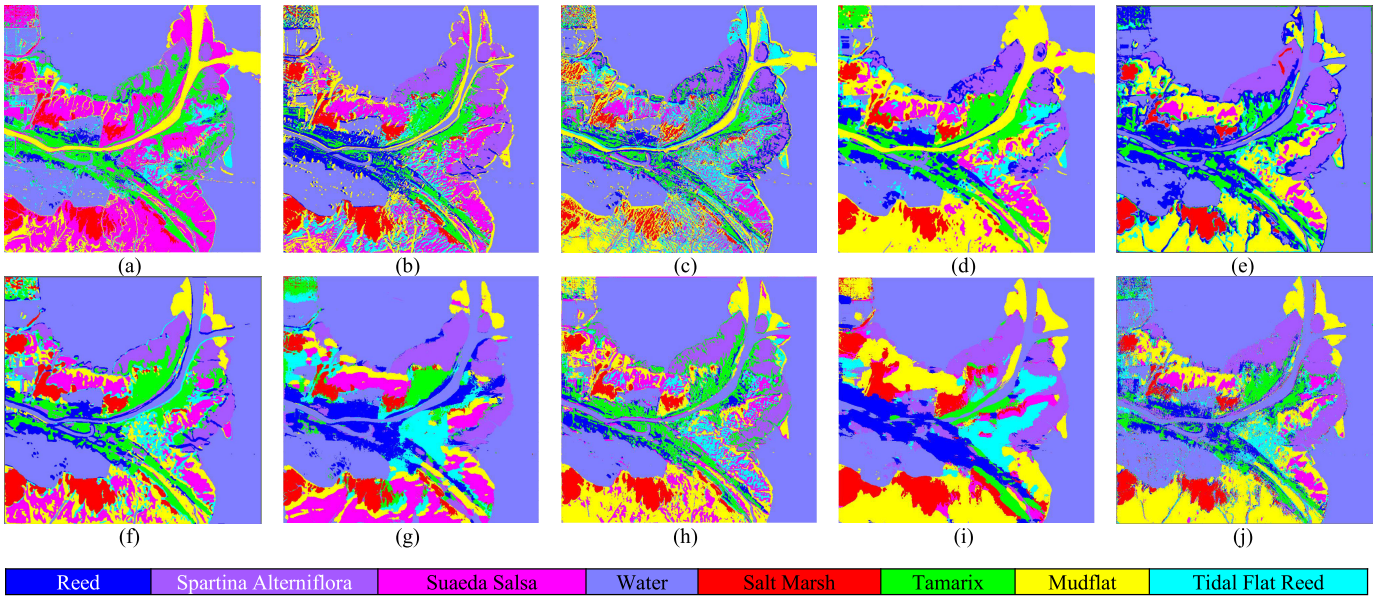


Fig. 7. Classification results on YRE_1 dataset. (a) SVM. (b) 3DCNN-H. (c) 3DCNN-M. (d) CLDA. (e) DCFSL. (f)CMFSL. (g) HFSL. (h) Gia-CFSL. (i) FSCF-SSL. (j) Ours method.

in Figs. 10, 11 and 12. It can be seen from Fig. 10(a) that in the original feature space, samples of different categories are mixed very messy, such as the Railway (class 11), Parking Lot1 (class 12), and Parking Lot 2 (class 13) categories in UH dataset. This shows that these classes are difficult to distinguish in the original feature space, and it can be seen from the UH dataset classification results in Table III that the classification results of categories 11, 12, and 13 in SVM are very poor. In YRE_1 and YRE_2 datasets, Reed and Tamarix (class 1 and class 2) also have obvious aliasing, so SVM is not effective in classifying these two types of ground objects, indicating that only spectral information in the original feature space cannot effectively distinguish different ground objects. In the CLDA method, knowledge transfer is achieved by adapting the source and target domains through adversarial learning. The t-SNE map of it is better than that of the original feature space. For example, compared with the original feature space map, Residential and Parking Lot1 in CLDA have less

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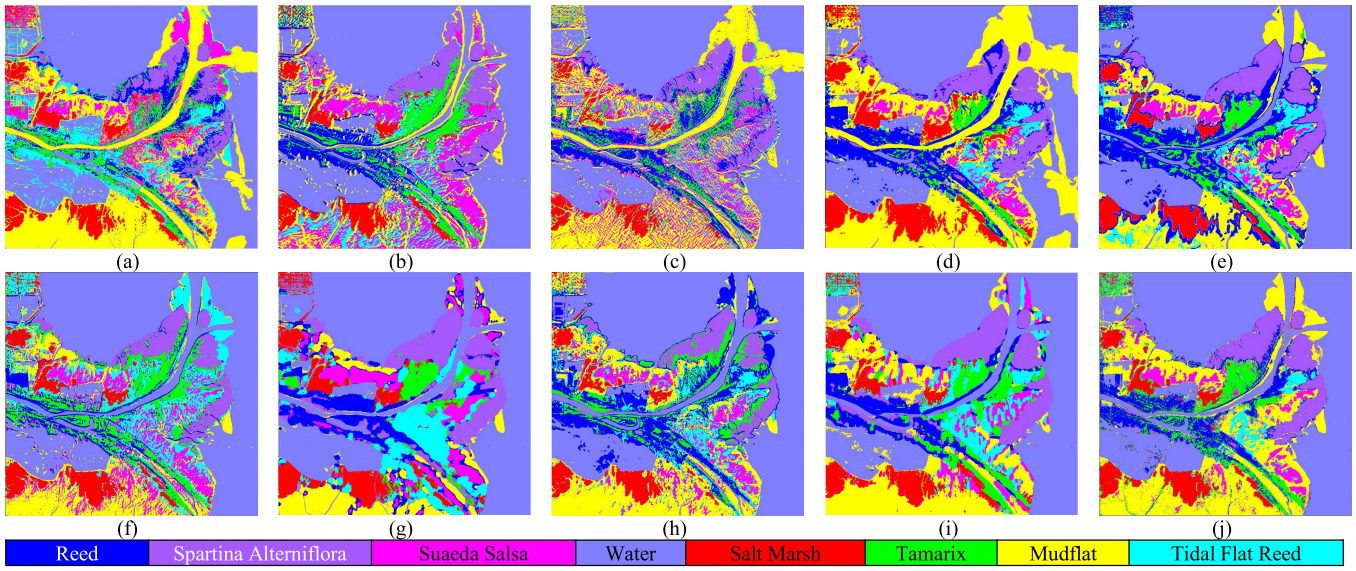


Fig. 8. Classification results on YRE_2 dataset. (a) SVM. (b) 3DCNN-H. (c) 3DCNN-M. (d) CLDA. (e) DCFSL. (f) CMFSL. (g) HFSL. (h) Gia-CFSL. (i) FSCF-SSL. (j) Ours method.

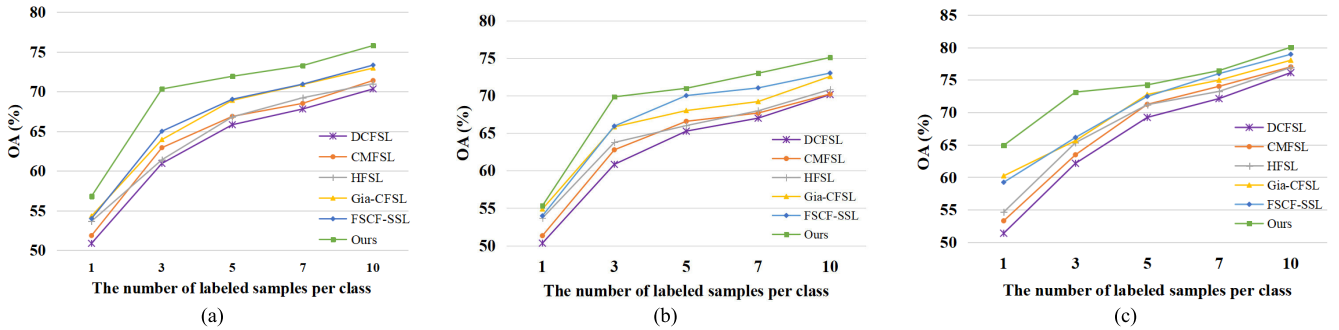


Fig. 9. Classification accuracies comparison of different methods with different numbers of labeled samples. (a) UH dataset. (b) YRE_1 dataset. (c) YRE_2 dataset.

TABLE VI

NUMBER OF TRAINING AND TESTING SAMPLES OF YRE_2 DATASET WITH SMALLER OVERLAPPING AREA

No.	Classes	TR in Source	TE
1	Reed	50	98275
2	Tamarix	50	55139
3	Tidal Flat Reed	35	43891
4	Salt Marsh	50	40126
5	Suaeda Salsa	50	53874
6	Mudflat	50	232701
7	Water	50	562243
8	Spartina Alterniflora	50	94479
	Total	385	1180728



TABLE VII

NUMBER OF TRAINING AND TESTING SAMPLES OF YRE_2 DATASET WITH BOUNDARY OVERLAPPING AREA

No.	Classes	TR in Source	TE
1	Reed	50	98275
2	Tamarix	50	55139
3	Tidal Flat Reed	50	43891
4	Salt Marsh	50	40126
5	Suaeda Salsa	50	53874
6	Mudflat	50	232701
7	Water	50	562243
8	Spartina Alterniflora	50	94479
	Total	400	1180728



confusion in the UH dataset, so the CLDA method achieves better classification results than SVM. In the cross-domain FSL methods, different ground feature features are dispersed through the adaptation of source and target domains and FSL, which increases the distance between classes and enhances the discriminability of features, so the classification effect is improved. Among these methods, HFSL and FSCF-SSL utilize natural images as the source domain, while other FSL methods use HS images as the source domain. However, it can be seen from the figures that compared with the proposed method, other cross-domain FSL methods do not gather the features of the same category together, and the features of the same category are scattered. The method we proposed can better integrate the same category in the dataset, such as the Commercial category in the UH dataset, and the Salt Marsh category in the Yellow River Estuary dataset. This shows that the method proposed in this article can learn a better metric space through sub-domain alignment, FDE and FBA, in which the same category features are condensed as much as possible.

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TABLE VIII
CLASSIFICATION RESULTS ON THE YRE_2 DATASET WITH SMALLER OVERLAPPING AREA

Methods	SVM	3DCNN-H	3DCNN-M	CLDA	DCFSL	CMFSL	HFSL	Gia-CFSL	FSCF-SSL	Ours
Reed	23.47	43.99	51.4	57.48	63.88	61.45	66.44	41.13	64.24	**67.02**
Tamarix	**63.21**	38.76	28.04	59.08	56.87	53.22	46.51	56.25	50.24	34.42
Tidal Flat Reed	20.47	26.39	20.44	36.52	31.88	37.71	30.88	24.38	**38.98**	25.37
Salt Marsh	91.79	97.49	96.41	**98.22**	97.24	95.54	95.28	85.15	97.92	94.04
Suaeda Salsa	46.39	**76.94**	52.45	37.32	47.41	45.66	52.97	62.88	31.34	46.07
Mudflat	61.6	55.91	61.7	52.63	41.28	45.82	48.86	69.81	56.79	**69.87**
Water	69.56	79.42	76.03	78.39	86.51	85.88	83.54	84.67	**87.81**	87.43
Spartina Alterniflora	54.24	74.18	65.43	79.93	75.38	74.70	76.23	77.69	74.39	**82.91**
OA	60.35	65.49	63.58	64.63	66.01	66.46	68.01	68.99	70.81	**71.76**
AA	53.84	61.64	56.49	62.44	62.49	62.93	62.58	62.63	62.71	**63.39**
Kappa	0.4873	0.5463	0.5206	0.542	0.5532	0.5572	0.5784	0.5901	0.608	**0.6193**

TABLE IX
CLASSIFICATION RESULTS ON THE YRE_2 DATASET WITH BOUNDARY OVERLAPPING AREA

Methods	SVM	3DCNN-H	3DCNN-M	CLDA	DCFSL	CMFSL	HFSL	Gia-CFSL	FSCF-SSL	Ours
Reed	39.79	41.91	25.76	27.84	31.63	66.8	39.53	56.85	**69.15**	67.53
Tamarix	**54.92**	51.76	42.94	30.03	65.92	43.9	58.24	37.69	51.01	37.02
Tidal Flat Reed	27.7	20.08	14.18	38.02	24.83	39.55	31.56	26.39	**40.35**	29.48
Salt Marsh	90.99	91.28	89.87	**97.47**	94.39	96.05	93.58	93.05	94.12	92.93
Suaeda Salsa	55.79	48.46	38.24	26.83	33.24	36.90	40.54	**60.11**	32.24	36.35
Mudflat	19.18	55.41	65.06	66.77	58.52	50.07	58.57	61.10	58.46	**68.50**
Water	89.26	89.32	78.69	79.57	89.21	88.60	88.78	87.46	86.04	**89.36**
Spartina Alterniflora	54.72	57.27	70.32	81.14	73.34	75.30	78.36	73.79	71.34	**81.74**
OA	63.37	69.98	66.31	68.83	70.98	71.09	71.42	72.97	73.11	**75.97**
AA	54.04	56.93	53.13	55.98	58.84	62.14	61.15	62.06	62.83	**62.86**
Kappa	0.5127	0.592	0.5514	0.582	0.6025	0.6125	0.6136	0.6297	0.6383	**0.667**

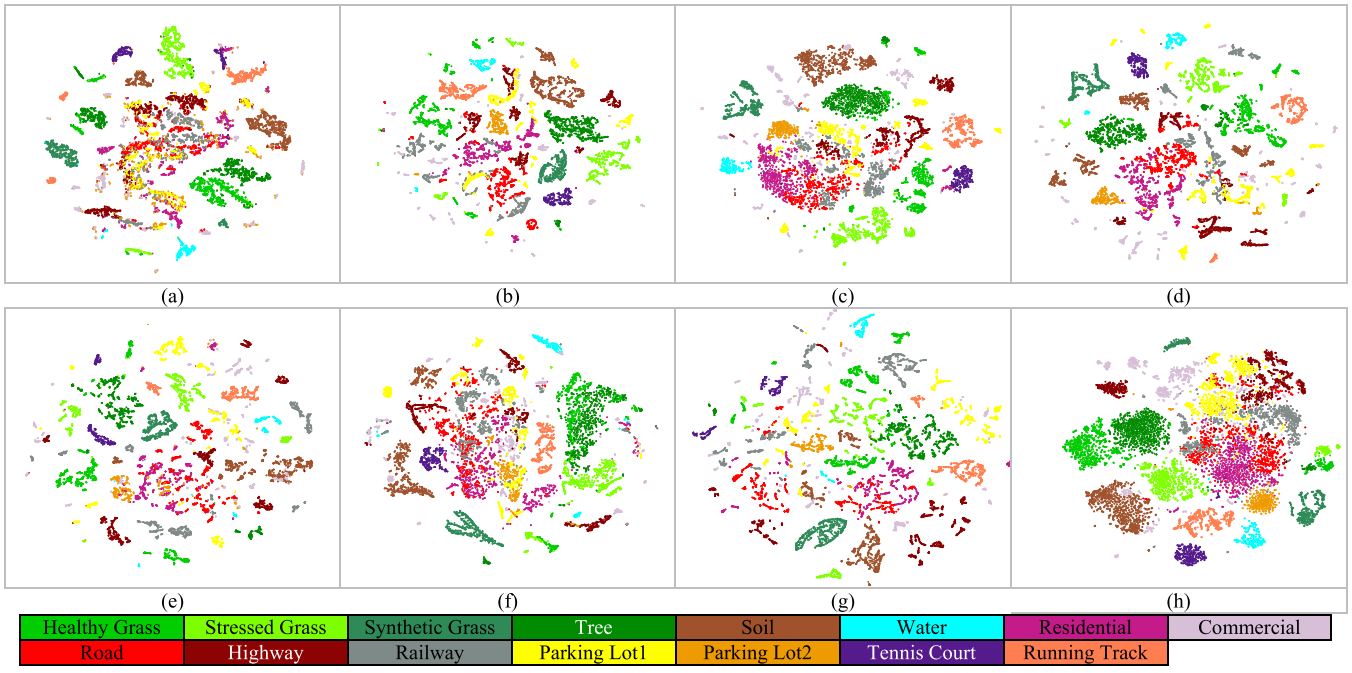


Fig. 10. *t*-SNE visualization on the UH dataset. (a) Original features. (b) CLDA. (c) DCFSL. (d) CMFSL. (e) HFSL. (f) Gia-CFSL. (g) FSCF-SSL. (h) Ours.

That is to say, the proposed method effectively reduces the intraclass distance and increases the interclass distance, so our method achieves the best classification effect.

E. Ablation Study

The framework proposed in this article is primarily composed of four main components: FSL, FDE module, subdomain alignment module, and FBA module. As shown in Table X, ablation analysis was conducted by removing each component, resulting in a total of five variants: ours-V0, representing the baseline network that retains only the original feature extraction part and FSL for the source and target domains; ours-V1, indicating that FSL is removed in the model; ours-V2, representing the deletion of subdomain

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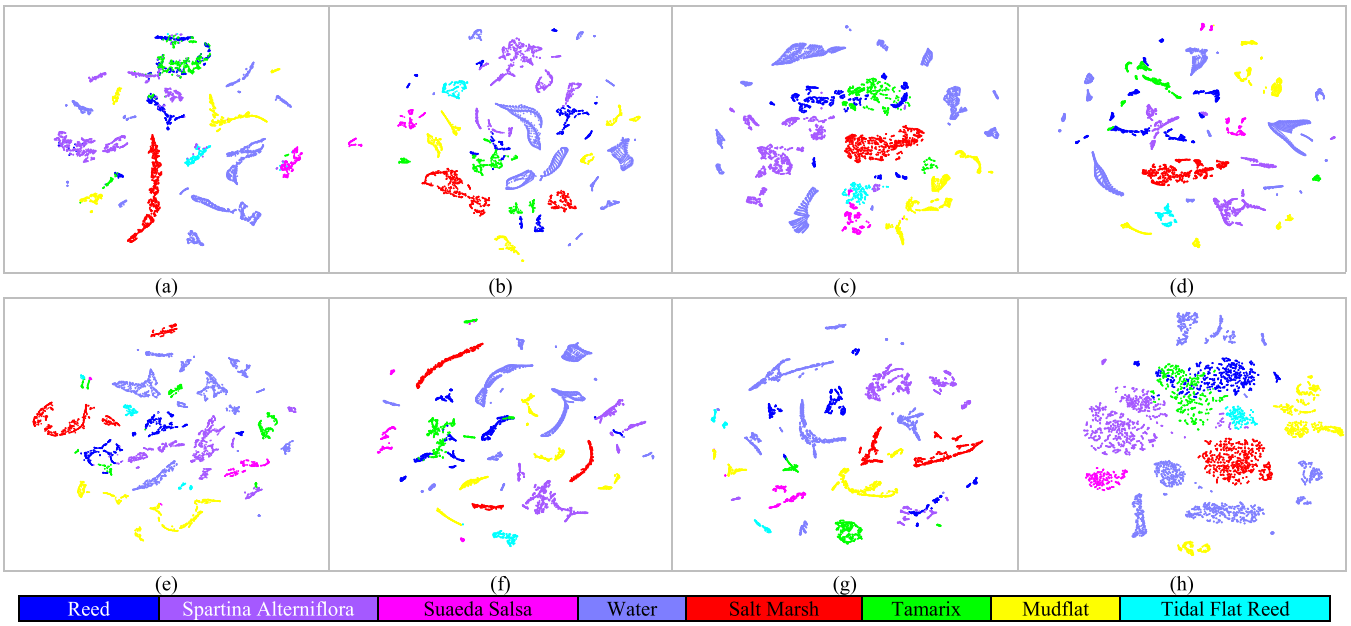


Fig. 11. t-SNE visualization on the YRE_1 dataset. (a) Original features. (b) CLDA. (c) DCFSL. (d) CMFSL. (e) HFSL. (f) Gia-CFSL. (g) FSCF-SSL. (h) Ours.

TABLE X
SET ARCHITECTURES WITH DIFFERENT MODULE

	FSL	FDE	SA	FBA
ours-V0	√			
ours-V1		√	√	√
ours-V2	√	√	√	
ours-V3	√	√		√
ours-V4	√		√	√
ours	√	√	√	√

alignment in the model; ours-V3, indicating that the FDE module is removed in the model; and ours-V4, representing the removal of the FBA module.

In this article, ten labeled samples were randomly selected from each target domain, and the experiment was independently repeated ten times. The mean and standard deviation of the experimental results of OA and kappa are listed in Table XI. As shown in the table, the proposed method outperforms other variants in terms of overall performance. Specifically, compared to ours-V1 and the overall framework ours, it can be observed that FSL in both the source and target domains, as well as the loss constraint of supervised classification, are critical in improving classification accuracy. Moreover, as shown by ours-V2 and ours, subdomain alignment plays a significant role in reducing the difference between the source and target domains, thereby enhancing the classification performance of the target domain. As demonstrated by ours-V4, the FDE is also essential in enhancing the diversity of sample features, improving the generalization ability of the model for multispectrum, and ultimately improving the accuracy rate. Comparing the overall method to ours-V5 reveals that the channel bias module can eliminate feature bias between the source and target domains, thereby improving the classification performance to some extent. Overall, the above modules all have a positive impact on improving the classification accuracy of the proposed model.

TABLE XI
ABLATION COMPARISON OF EACH VARIANT

	ours-V0	ours-V1	ours-V2	ours-V3	ours-V4	ours
Dataset	University of Houston					
OA	75.62 ±2.13	58.09 ±1.92	79.02 ±1.18	77.55 ±1.61	78.32 ±1.56	80.04 ±1.04
kappa	73.68 ±2.32	54.7 ±2.01	77.33 ±1.27	75.76 ±1.74	76.53 ±1.68	78.43 ±1.13
Dataset	YELLOW RIVER ESTUARY_1					
OA	69.87 ±3.12	56.21 ±2.62	74.69 ±1.51	72.95 ±1.84	73.23 ±1.64	75.80 ±1.43
kappa	60.32 ±3.79	43.92 ±2.81	64.68 ±1.69	63.37 ±2.03	63.64 ±1.93	66.96 ±1.67
Dataset	YELLOW RIVER ESTUARY_2					
OA	69.17 ±3.23	54.19 ±2.05	74.12 ±1.62	72.21 ±1.96	73.03 ±1.78	75.09 ±1.54
kappa	59.03 ±3.91	42.13 ±2.36	64.78 ±1.89	62.48 ±2.19	63.44 ±2.07	65.61 ±1.86

What is more, in order to further verify the effect of the FBA module, the t-SNE map is employed to visualize the source domain and target domain features of ours and ours-V4 methods on three datasets respectively. Among them, the HS image features of the source domain training data are denoted by "+," and the MS image features of the target domain test data are represented by "·." As illustrated in Figs. 13 and 14, a noticeable feature offset is observed in the Residential class in the UH dataset. Our method significantly reduces this feature offset between the Residential class in the source domain and the corresponding class in the target domain. In the two sets of Yellow River Estuary datasets, there is also an obvious shift after the feature mapping of the Salt Marsh class in the source domain and target domain in the ours-V4 method, and the feature shift is reduced in our method. Compared with the ours-V4 method, the feature offset between the source domain and the target domain in our method is smaller, and features of the same category are

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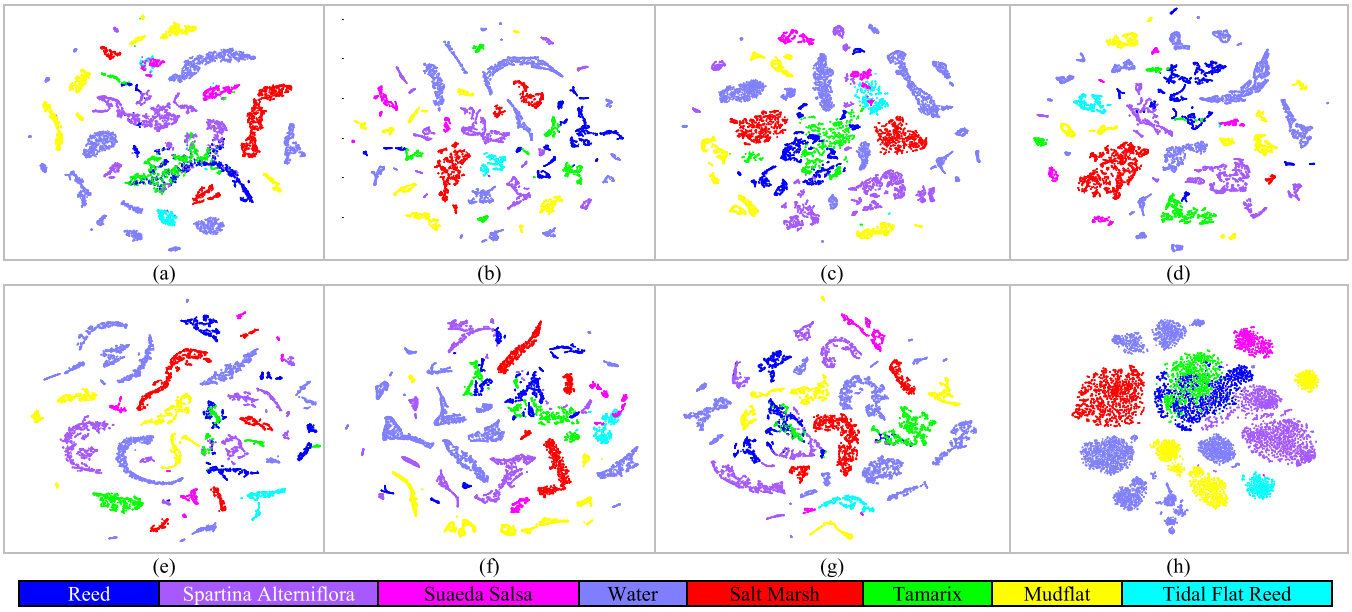


Fig. 12. t-SNE visualization on the YRE_2 dataset. (a) Original features. (b) CLDA. (c) DCFSL. (d) CMFSL. (e) HFSL. (f) Gia-CFSL. (g) FSCF-SSL. (h) Ours.

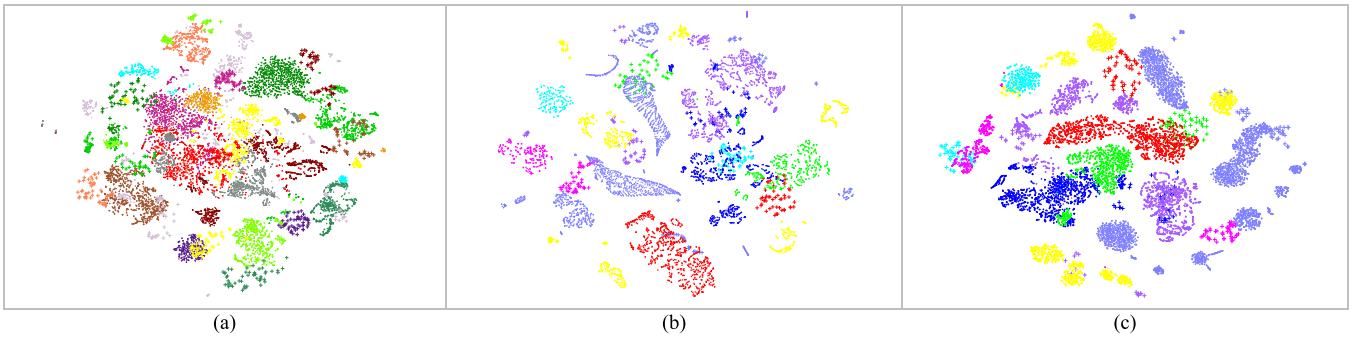


Fig. 13. t-SNE visualization of ours-V4 on three datasets. (a) The UH dataset. (b) The YRE_1 dataset. (c) The YRE_2 dataset.

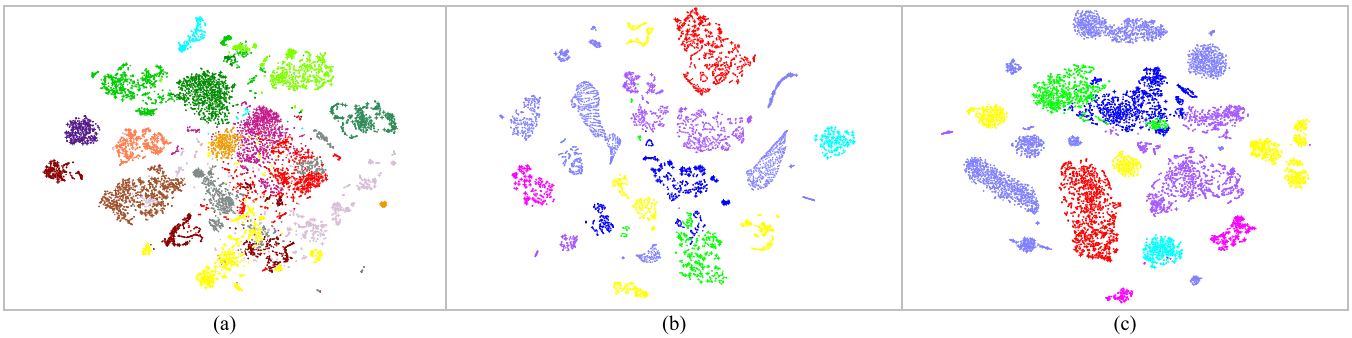


Fig. 14. t-SNE visualization of ours on three datasets. (a) The UH dataset. (b) The YRE_1 dataset. (c) The YRE_2 dataset.

clustered together after mapping. Therefore, the FBA module can reduce the feature shift between the source domain and the target domain, thereby improving the classification ability of the target domain.

F. Computational Complexity Analysis

To analyze the complexity of the proposed method, we compare the running time of the proposed method in floating point operations (FLOPs) and the number of parameters with cross-domain small sample comparison algorithms under ten labeled samples in three datasets. All methods are executed on NVIDIA Tesla V100 GPU, and the detailed statistical results are shown in Table XII. Since DCFSL uses a fully connected layer for domain adaptation, there are a large number of parameters. Compared with the cross-domain FSL methods, our method requires fewer parameters and less computation. This is mainly due to the proposed method having fewer parameters and less computation due to the proposed sub-domain alignment algorithm. The proposed method helps the

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TABLE XII
RUNNING TIME (S), FLOPS (M, MILLION), AND NUMBER OF PARAMETERS (M, MILLION) OF DIFFERENT METHODS

Datasets	Metrics	DCFSL	CMFSL	HFSL	Gia-CFSL	FSCF-SSL	Ours
University of Houston	Training time(s)	1032	1390	2041	2979	12264	1497
	Test time (s)	6.07	3.19	9.31	2.77	9.51	2.03
	FLOPs（MMac）	42.91	25.89	199.63	42.61	397.32	13.39
	Params（M）	4.27	0.50	3.08	0.45	3.08	0.16
Yellow River Estuary_1	Training time (s)	2643	2057	3796	3192	15341	2498
	Test time (s)	7.76	4.38	10.63	4.01	11.65	3.26
	FLOPs（MMac）	43.63	27.11	200.49	43.14	399.92	13.81
	Params（M）	4.28	0.51	3.1	0.46	3.1	0.17
Yellow River Estuary_2	Training time (s)	2743	2063	3806	3284	16982	2501
	Test time (s)	8.06	5.46	11.43	4.21	11.74	3.39
	FLOPs（MMac）	43.63	27.11	200.49	43.14	399.92	13.81
	Params（M）	4.28	0.51	3.1	0.46	3.1	0.17

model to transfer enough knowledge from the source class to the target class, so as to achieve better results. The proposed method has the shortest prediction period for the classification results of the three datasets. Therefore, the proposed method has certain advantages in both classification accuracy and model complexity.

IV. CONCLUSION

In this article, a new MS-HS collaborative classification method is proposed. The main purpose is to transfer the excellent spectral resolution of HS images to MS images and improve the classification of large scene MS images. First of all, due to the low feature diversity of small sample categories, this method proposes a 3-D FDE module to simulate various feature distributions of samples in the training stage to improve the generalization ability of the model. Second, the source domain and the target domain are adapted by means of subdomain alignment, so that the learned metric space has a large intraclass distance and a small interclass distance, reducing the classification error. Finally, in the testing phase, the method introduces the FBA module, which further reduces the bias between the source domain and the target domain through a simple channel-by-channel feature transformation and improves the classification ability. The experiments demonstrate that the proposed method achieves superior performance even with limited samples. In future work, we will study the situation where the source domain and the target domain are more different, such as the cooperative use of HS images and SAR images.

REFERENCES

[1] L. He, S.-L. Qi, J.-Z. Duan, T.-C. Guo, W. Feng, and D.-X. He, "Monitoring of wheat powdery mildew disease severity using multiangle hyperspectral remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 979–990, Feb. 2021.

[2] Y. Gu, J. Chanussot, X. Jia, and J. A. Benediktsson, "Multiple kernel learning for hyperspectral image classification: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6547–6565, Nov. 2017.

[3] X. Li, M. Ding, Y. Gu, and A. Pižurica, "An end-to-end framework for joint denoising and classification of hyperspectral images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 7, pp. 3269–3283, Jul. 2023.

[4] X. Lu, W. Zhang, and X. Li, "A hybrid sparsity and distance-based discrimination detector for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1704–1717, Mar. 2018.

[5] X. Zheng, X. Chen, X. Lu, and B. Sun, "Unsupervised change detection by cross-resolution difference learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5606616.

[6] M. Shimoni, R. Haelterman, and C. Perneel, "Hypersectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[7] I. Makki, R. Younes, C. Francis, T. Bianchi, and M. Zucchetti, "A survey of landmine detection using hyperspectral imaging," *ISPRS J. Photogramm. Remote Sens.*, vol. 124, pp. 40–53, Feb. 2017.

[8] Y. Gu et al., "Multimodal hyperspectral remote sensing: An overview and perspective," *Sci. China Inf. Sci.*, vol. 64, Feb. 2021, Art. no. 121301.

[9] S. Shekhar, V. M. Patel, H. V. Nguyen, and R. Chellappa, "Generalized domain-adaptive dictionaries," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 361–368.

[10] Q. Tian, H. Sun, C. Ma, M. Cao, Y. Chu, and S. Chen, "Heterogeneous domain adaptation with structure and classification space alignment," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 10328–10338, Oct. 2022.

[11] Y. Zhu et al., "Deep subdomain adaptation network for image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 4, pp. 1713–1722, Apr. 2021.

[12] H. Li, S. J. Pan, S. Wang, and A. C. Kot, "Heterogeneous domain adaptation via nonlinear matrix factorization," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 3, pp. 984–996, Mar. 2020.

[13] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "CoSpace: Common subspace learning from hyperspectral-multispectral correspondences," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4349–4359, Jul. 2019.

[14] B. Guo, T. Liu, and Y. Gu, "Integrating coupled dictionary learning and distance preserved probability distribution adaptation for multispectral–hyperspectral image collaborative classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528013.

[15] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[16] D. Hong, J. Yao, D. Meng, Z. Xu, and J. Chanussot, "Multimodal GANs: Toward crossmodal hyperspectral–multispectral image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5103–5113, Jun. 2021.

[17] C. Zhao, B. Qin, S. Feng, W. Zhu, L. Zhang, and J. Ren, "An unsupervised domain adaptation method towards multi-level features and decision boundaries for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5546216.

[18] B. Guo, T. Liu, and Y. Gu, "Structure preserved discriminative distribution adaptation for multihyperspectral image collaborative classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5526815.

[19] D. Hong, N. Yokoya, G.-S. Xia, J. Chanussot, and X. X. Zhu, "X-ModalNet: A semi-supervised deep cross-modal network for classification of remote sensing data," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 12–23, Sep. 2020.

[20] Z. Fang et al., "Confident learning-based domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527116.

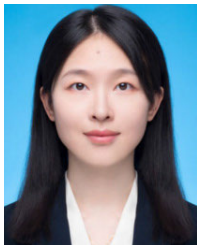
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[21] S. Mei, J. Ji, J. Hou, X. Li, and Q. Du, "Learning sensor-specific spatial–spectral features of hyperspectral images via convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4520–4533, Aug. 2017.
[22] J. Yang, Y.-Q. Zhao, and J. C. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.
[23] O. Vinyals, C. Blundell, T. Lillicrap, and D. Wierstra, "Matching networks for one shot learning," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2016, pp. 1–9.
[24] Q. Sun, Y. Liu, T.-S. Chua, and B. Schiele, "Meta-transfer learning for few-shot learning," in *Proc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 403–412.
[25] H.-Y. Tseng, H.-Y. Lee, J.-B. Huang, and M.-H. Yang, "Cross-domain few-shot classification via learned feature-wise transformation," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2020, pp. 1–18.
[26] X. Luo, J. Xu, and Z. Xu, "Channel importance matters in few-shot image classification," in *Proc. Int. Conf. Mach. Learn.*, Jun. 2022, pp. 14542–14559.
[27] A. Santoro, S. Bartunov, M. Botvinick, D. Wierstra, and T. Lillicrap, "Meta-learning with memory-augmented neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1842–1850.
[28] T. Elsken, B. Staffler, J. H. Metzen, and F. Hutter, "Meta-learning of neural architectures for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12362–12372.
[29] F. Sung, Y. Yang, L. Zhang, T. Xiang, P. H. S. Torr, and T. M. Hospedales, "Learning to compare: Relation network for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1199–1208.
[30] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for fewshot learning," in *Proc. Neural Inf. Process. Syst., Mar.*, 2017, pp. 1–11.
[31] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.
[32] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.
[33] J. Yang, H. Zou, Y. Zhou, Z. Zeng, and L. Xie, "Mind the discriminability: Asymmetric adversarial domain adaptation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 589–606.
[34] C. Yu, B. Gong, M. Song, E. Zhao, and C.-I. Chang, "Multi-view calibrated prototype learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5544713.
[35] Y. Wang et al., "Heterogeneous few-shot learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[36] Z. Li, H. Guo, Y. Chen, C. Liu, Q. Du, and Z. Fang, "Few-shot hyperspectral image classification with self-supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5517917.
[37] Q. Liu et al., "Refined prototypical contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506214.
[38] C. Yu, J. Huang, M. Song, Y. Wang, and C.-I. Chang, "Edge-inferring graph neural network with dynamic task-guided self-diagnosis for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535613.
[39] H. Wang, X. Wang, and Y. Cheng, "Graph meta transfer network for heterogeneous few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 202, Art. no. 5501112.
[40] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, 2022, doi: 10.1109/TNNLS.2022.3185795.
[41] T. Wang, H. Liu, and J. Li, "Spectral–spatial classification of few shot hyperspectral image with deep 3-D convolutional random Fourier features network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532318.
[42] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022.
[43] J. Sun, X. Shen, and Q. Sun, "Hyperspectral image few-shot classification network based on the Earth Mover's distance," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5534114.



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